CS23334 Fundamentals of Data Science Lab

Experiment 3.a

Conduct an experiment to show data visualization using line plot

Description: Take any sample data either through csv file or data fetched directly through code.

Code:

```
import matplotlib.pyplot as cricket

Overs=list(range(5,51,5))

Indian_Score=[30,55,90,129,165,200,239,270,310,350]

Srilankan_Score=[25,70,90,120,140,170,195,220,255,279]

cricket.plot(Overs,Indian_Score)

cricket.plot(Overs,Srilankan_Score)

cricket.show()

cricket.title("INDIA Vs SRILANKA")

cricket.xlabel("Overs")

cricket.ylabel("Score")

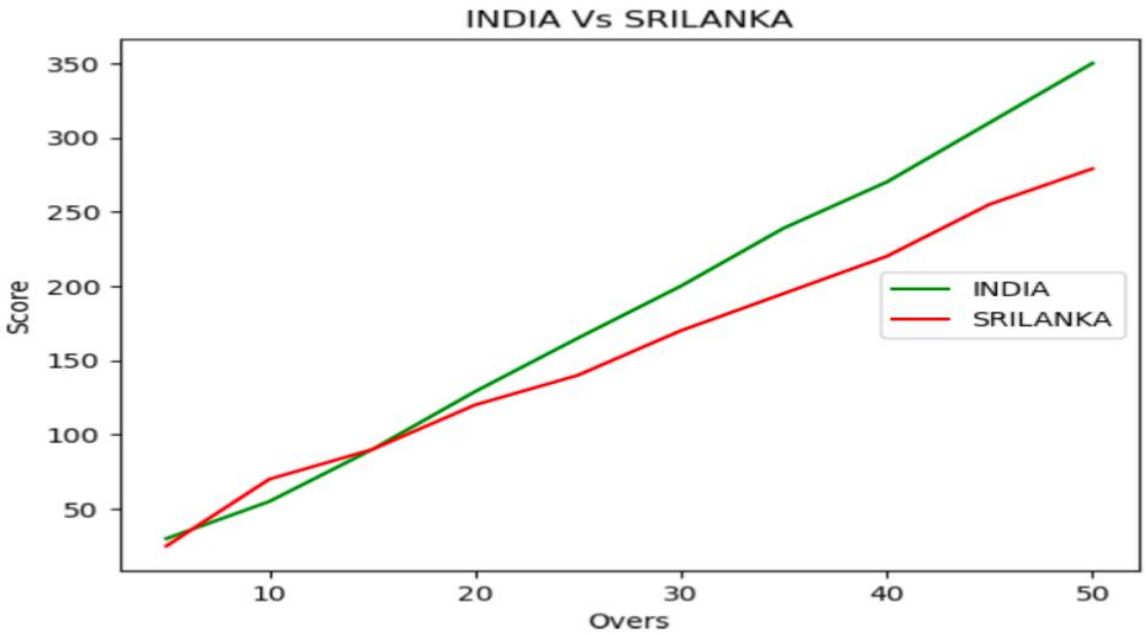
cricket.legend()

cricket.plot(Overs,Indian_Score,color="green",label="INDIA")

cricket.plot(Overs,Srilankan_Score,color="red",label="SRILANKA")

cricket.legend(loc="center right")
```

OUTPUT:


INDIA Vs SRILANKA
350
300
250
200
150
100
50
Score
10
20
30
40
50
Overs
INDIA
SRILANKA

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Experiment 3.b

Conduct an experiment to show data visualization using bar chart

Description: Take any sample data either through csv file or data fetched directly through code.

Code:

```
import matplotlib.pyplot as hscmark

import numpy as np

Names = ['SHREE', 'DEV', 'KEERTHI', 'PRIYA', 'SHAN', 'KUMARAN']

xaxis = np.arange(len(Names))

Percentage_hsc = [96, 91, 94, 75, 45, 81]

hscmark.bar(Names, Percentage_hsc)

hscmark.xticks(xaxis, Names, rotation=45)

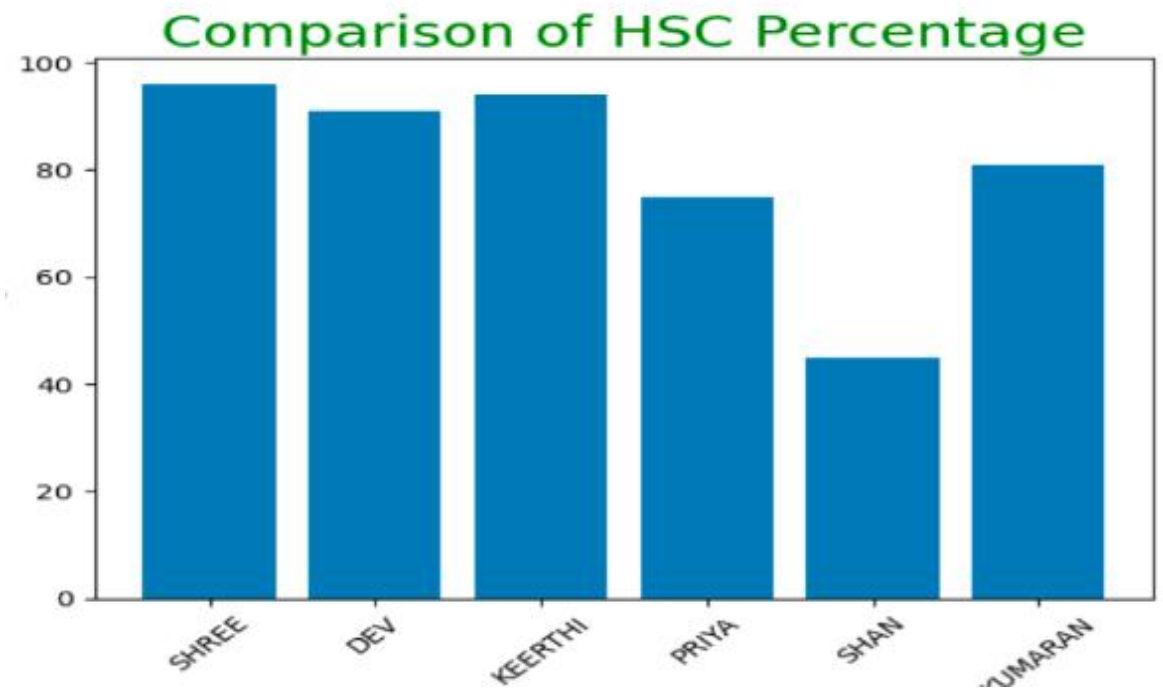
hscmark.xlabel("Names of Pupil")

hscmark.ylabel("Percentage")

hscmark.title("Comparison of HSC Percentage", fontsize=20, color="green")

hscmark.show()
```

OUTPUT:


Comparison of HSC Percentage
100
80
60
40
20
0
SHREE
DEV
KEERTHI
PRIYA
SHAN

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Experiment 3.c

Conduct an experiment to show data visualization using pie chart

Description: Take any sample data either through csv file or data fetched directly through code.

Code:

```
import matplotlib.pyplot as election

labels = ['CANDIDATE 1', 'CANDIDATE 2', 'CANDIDATE 3', 'CANDIDATE 4']

Votes = [315, 130, 245, 210]

colors = ['green', 'yellow', 'red', 'orange']

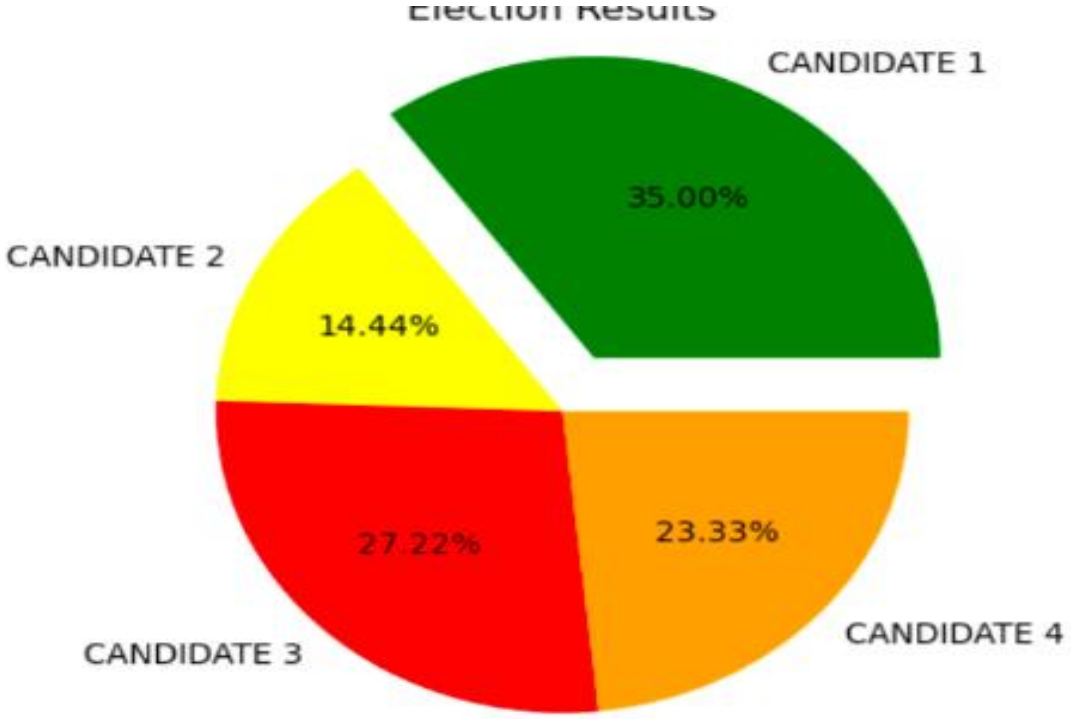
explode = (0.2, 0, 0, 0)

election.pie(Votes, labels=labels, colors=colors, explode=explode, autopct='%0.2f%%')

election.title('Election Results')

election.show()
```

OUTPUT:



Result:

Line plots were successfully generated for the India vs Sri Lanka cricket match scorecard. The visualization clearly depicted India's higher run rate across all overs, with both teams' scoring trends plotted on the same axes for easy comparison.